UNIT V – OPERATIONAL & INTEGRATED RISK MANAGEMENT

Financial risk does not arise from only one source. A bank, company, financial institution or government may face **operational risk, credit risk, counterparty risk, systemic risk and sovereign risk at the same time**.

Integrated Risk Management means managing all major risks together rather than looking at each risk separately.

Simple example

Suppose an Indian company imports machinery from Germany.

It may face:

- **Credit risk** – the German supplier may fail to perform.
- **Counterparty risk** – the bank providing a foreign-exchange contract may fail.
- **Operational risk** – an employee may enter the wrong transaction.
- **Foreign exchange risk** – the euro may become expensive.
- **Sovereign risk** – government restrictions may affect international payments.
- **Systemic risk** – a banking or financial crisis may disrupt the entire market.

Therefore, risk management should consider the **combined effect of different risks**.

OPERATIONAL RISK

Meaning of Operational Risk

Operational risk is the risk of loss arising from inadequate or failed:

- internal processes,
- people,
- systems,
- technology,
- or external events.

Simple definition

Operational risk is the risk of loss caused by something going wrong in the way an organisation operates.

Examples

- Employee enters an incorrect payment amount.

- Bank's computer system crashes.
- Fraud is committed by an employee.
- Cyberattack causes a data breach.
- Fire damages an office.
- Important documents are lost.
- A company follows an incorrect process.
- A supplier fails to deliver an important service.

3. Sources of Operational Risk

Operational risk can broadly arise from four major sources.

People

Problems caused by employees or other individuals.

Examples:

- Employee fraud
- Lack of training
- Human error
- Poor decision-making
- Employee negligence

Processes

Failures in internal procedures.

Examples:

- Incorrect approval procedure
- Incomplete documentation
- Poor internal controls
- Failure to follow compliance procedures

Systems

Failures in technology or information systems.

Examples:

- Software failure
- Network failure

- Server breakdown
- Cyberattack
- Incorrect programming

External Events

Events outside the organisation.

Examples:

- Natural disasters
- Fire
- Pandemic
- Political disturbance
- Terrorist attack
- Regulatory changes

Measuring Operational Risk

Operational risk is difficult to measure because losses may occur irregularly and may not follow a predictable pattern.

Organisations generally use both **qualitative and quantitative approaches**.

Qualitative Measurement

This involves evaluating risk using:

- expert judgement,
- risk assessment,
- interviews,
- checklists,
- workshops,
- SWOT analysis,
- brainstorming,
- root cause analysis.

Example

A bank may identify:

"There is a high possibility of fraudulent online transactions."

Management then evaluates:

- probability of fraud,
- possible financial loss,
- existing controls,
- additional controls required.

Quantitative Measurement

Quantitative methods use numerical information.

Examples include:

- frequency of losses,
- average loss,
- maximum historical loss,
- expected loss,
- Value at Risk-type measures,
- scenario analysis,
- stress testing.

Basic illustration

Suppose a company experienced operational losses:

Year	**Loss**
2023	₹2 lakh
2024	₹4 lakh
2025	₹3 lakh
2026	₹6 lakh

Average loss:

$$\frac{2+4+3+6}{4} = ₹3.75 \text{ lakh}$$

The company may use this historical information as one input for operational-risk planning.

Operational Risk Management

Operational risk management involves:

Identify → Assess → Control → Monitor → Report

Step 1 – Identify

Find possible sources of operational risk.

Step 2 – Assess

Estimate:

- probability of occurrence,
- financial impact,
- reputational impact,
- regulatory impact.

Step 3 – Control

Introduce controls to reduce risk.

Step 4 – Monitor

Continuously observe whether controls are working.

Step 5 – Report

Report significant risks and losses to management.

Risk Control Techniques

Organisations can manage operational risk through:

- **Avoidance**

Stop an activity that has unacceptable risk.

Example: A company stops using an unreliable payment platform.

- **Reduction**

Reduce the probability or impact of the risk.

Example: Introduce two-factor authentication.

- **Transfer**

Transfer the financial consequences to another party.

Example: Insurance.

- **Acceptance**

Accept the risk when the cost of controlling it is greater than the expected loss.

7. BRAINSTORMING FOR RISK IDENTIFICATION

Meaning

Brainstorming is a group technique used to identify as many possible risks as possible.

Participants freely suggest ideas without immediately criticising them.

Example

A bank wants to identify risks associated with online banking.

Participants may identify:

- cyberattack,
- password theft,
- system failure,
- payment fraud,
- phishing,
- server failure,
- customer identity theft,
- employee misuse.

After brainstorming, the risks are classified and evaluated.

Advantages

- Simple to use
- Generates many ideas
- Encourages employee participation
- Useful for new projects
- Helps identify risks that may otherwise be ignored

Limitation

- Dominant individuals may influence the discussion.
- Some risks may be overlooked.
- Results depend on participant knowledge.

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8. SWOT ANALYSIS

Meaning

SWOT stands for:

- **S – Strengths**
- **W – Weaknesses**
- **O – Opportunities**
- **T – Threats**

It is used to analyse the internal and external environment of an organisation.

Internal factors

- Strengths
- Weaknesses

External factors

- Opportunities
- Threats

Example: Bank Introducing Digital Banking

SWOT	**Example**
Strength	Strong customer base
Weakness	Outdated IT infrastructure
Opportunity	Growing digital payments
Threat	Cyberattacks

Risk management use

SWOT helps management identify areas where risk may arise.

For example:

Weak IT infrastructure + increasing cyber threats = high technology risk.

ROOT CAUSE ANALYSIS

Meaning

Root Cause Analysis (RCA) is a technique used to identify the fundamental reason behind a problem.

Instead of asking:

"What happened?"

RCA asks:

"Why did it happen?"

Example

Problem

A bank transferred ₹50 lakh to the wrong account.

Why?

Employee entered an incorrect account number.

Why?

The employee manually entered the number.

Why?

There was no automated validation system.

Root cause

Weak transaction-control process.

The solution should therefore not simply be:

"Tell the employee to be careful."

Instead, the bank can introduce:

- automated account validation,
- maker-checker system,

- transaction limits,
- confirmation procedures.

Five Whys Technique

The **Five Whys** technique repeatedly asks "Why?" until the underlying cause is identified.

Example

Problem: Online payment failed.

1. Why? – Payment server was unavailable.
2. Why? – Server crashed.
3. Why? – Server was overloaded.
4. Why? – Capacity was insufficient.
5. Why? – Capacity planning was inadequate.

Root cause

Poor capacity planning.

ASSUMPTION ANALYSIS

Meaning

Assumption analysis examines the assumptions on which a business decision, project or risk assessment is based.

Many financial decisions are based on assumptions.

Examples

A company may assume:

- interest rates will remain stable,
- sales will increase by 10%,
- exchange rates will remain within a range,
- customers will repay loans on time,
- inflation will remain low.

If these assumptions prove wrong, the financial outcome may change significantly.

Example

A company plans a project based on:

Expected sales growth = 15%

But actual growth becomes only 5%.

The project's profitability may fall significantly.

Therefore, management should ask:

"What happens if our assumption is wrong?"

This leads to **scenario analysis and stress testing**.

CREDIT RISK

Meaning

Credit risk is the risk that a borrower will fail to meet its financial obligations.

Example

A bank lends ₹10 crore to Company A.

If Company A fails to repay the loan and interest, the bank suffers a loss.

That is **credit risk**.

Measuring Credit Risk

Credit risk can be assessed using:

- credit ratings,
- financial ratios,
- repayment history,
- cash-flow analysis,
- collateral,
- probability of default,
- exposure at default,
- loss given default.

Three important concepts are:

Probability of Default – PD

Probability that the borrower will fail to meet its obligations.

Loss Given Default – LGD

Percentage of exposure that the lender expects to lose if default occurs.

Exposure at Default – EAD

Amount exposed to loss when default occurs.

A simplified expected loss formula is:

$$EL = PD \times LGD \times EAD$$

Example

Suppose:

- PD = 5%
- EAD = ₹10 crore
- LGD = 40%

$$EL = 5\% \times 40\% \times ₹10 \text{ crore}$$
$$= ₹0.20 \text{ crore}$$

Therefore:

Expected Loss = ₹20 lakh

COUNTERPARTY RISK

Meaning

Counterparty risk is the risk that the other party to a financial contract fails to fulfil its contractual obligation.

It is particularly important in:

- derivatives,
- swaps,
- forwards,
- securities transactions,
- foreign exchange transactions.

Example

Company A enters into an interest-rate swap with Bank B.

If Bank B becomes unable to meet its obligation, Company A faces **counterparty risk**.

CREDIT RISK VS COUNTERPARTY RISK

Credit Risk	Counterparty Risk
Usually associated with borrowing/lending	Common in financial contracts
Borrower may fail to repay	Counterparty may fail to perform
Example: bank loan	Example: swap contract
Focuses on borrower creditworthiness	Focuses on contractual counterparty

SYNDICATION OF CREDIT

Meaning

Credit syndication occurs when several financial institutions jointly provide a large loan to one borrower.

It is generally used when the financing requirement is too large for one bank or when banks want to share risk.

Example

A company requires:

₹1,000 crore

Instead of one bank providing the entire amount:

- Bank A – ₹300 crore
- Bank B – ₹250 crore
- Bank C – ₹200 crore
- Bank D – ₹150 crore
- Bank E – ₹100 crore

Total = **₹1,000 crore**

This is a **syndicated loan**.

Advantages

For borrower

- Access to large funds
- Diversification of funding sources
- Can obtain competitive terms

For banks

- Risk sharing
- Reduced concentration
- Ability to participate in large transactions

NETTING OF PAYMENTS

Meaning

Netting means combining multiple obligations between parties and settling only the net amount.

Simple example

Company A owes Bank B:

₹10 crore

Bank B owes Company A:

₹7 crore

Instead of making two separate payments:

- A pays ₹10 crore
- B pays ₹7 crore

The net amount is:

$$₹10crore - ₹7crore = ₹3crore$$

Therefore, only **₹3 crore** needs to be settled.

Benefits of Netting

- Reduces settlement risk
- Reduces transaction costs
- Reduces cash requirements
- Reduces operational burden
- Reduces counterparty exposure

CROSS-COLLATERALIZATION

Meaning

Cross-collateralization means using the same collateral to secure more than one obligation or loan, subject to the contractual and legal arrangement.

Example

A business has two loans from a bank.

- Loan 1 = ₹5 crore
- Loan 2 = ₹3 crore

A property worth ₹10 crore is provided as security for both loans.

The same asset supports multiple obligations.

Benefit

It may provide greater security to the lender.

Risk

If the borrower defaults, the lender may have claims over the collateral for multiple obligations, depending on the agreement.

THIRD-PARTY GUARANTEE

Meaning

A third-party guarantee occurs when another party promises to meet the borrower's obligation if the borrower fails to do so.

Example

Company A wants a ₹20 crore loan.

Bank asks for a guarantee from Company B.

Company B agrees:

"If Company A fails to repay, I will meet the guaranteed obligation according to the terms of the guarantee."

Company B is the **guarantor**.

BANK GUARANTEE

A **bank guarantee** is a commitment by a bank to make payment to a beneficiary if its customer fails to fulfil specified obligations.

Example

A construction company receives a government contract.

The government requires a performance guarantee.

The company's bank issues a bank guarantee.

If the company fails to fulfil its contractual obligations, the beneficiary may invoke the guarantee subject to its terms.

CONTAGION

Meaning

Contagion means the spread of financial problems from one entity or market to another.

Example

A bank in Country A experiences severe financial problems.

Investors become nervous about banks in Country B.

They withdraw funds from those banks.

The problem spreads.

This is financial contagion.

SOVEREIGN RISK

Meaning

Sovereign risk is the risk that a country's government may be unable or unwilling to meet its financial obligations or may take actions that adversely affect investors.

Examples

- Government default
- Debt restructuring
- Capital controls
- Currency restrictions
- Political instability
- Sudden tax or regulatory changes

Measuring SOVEREIGN RISK

Sovereign risk can be assessed through:

- government debt,
- Debt/GDP ratio,
- Debt/Revenue ratio,
- debt-servicing burden,
- fiscal deficit,
- current account position,
- Balance of Payments,
- foreign exchange reserves,
- economic growth,
- inflation,
- political stability,
- demographic profile.